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High Performance Processors and Systems

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Name	
Last Name	

Problem 1 (25%)	
Problem 2 (25%)	
Question 1 (25%)	
Question 2 (25%)	
Total (100%)	
Extra (3 points)	
Total (x% + 3 points)	

Problem 1

Assume that the following code has been executed on a CPU with SCOREBOARD.

	Issue	Read Op	Exec Co.	Write R.
LD F6 32+ R2	1	2	3	4
LD F2 45+ R3	5	6	7	8
MULTD F0 F4 F2	6	9	19	20
ADD F2 F8 F6	8	9	10	11
DIVD F12 F0 F6	7	21	31	32
SUBD F6 F8 F2	9	10	11	12

A. Is there a “configuration” that can respect the shown execution?
How many units? Which kind? What latency?

B. If the previous table was not correct, please, write the right one and specify the number, kind and latency for each unit.

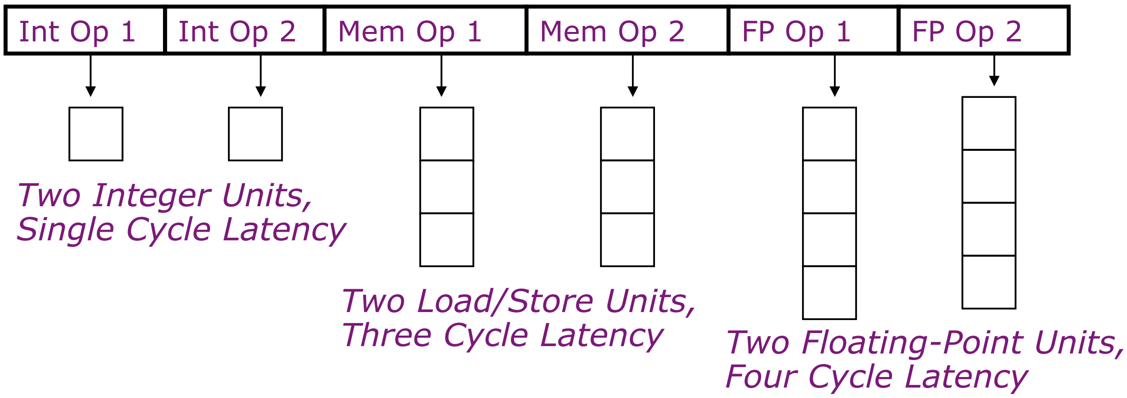
Answer. A

Answer. B

	Issue	Read Op	Exec Co.	Write R.
LD F6 32+ R2				
LD F2 45+ R3				
MULTD F0 F4 F2				
ADD F2 F8 F6				
DIVD F12 F0 F6				
SUBD F6 F8 F2				

Problem 2

Considering the following VLIW "architecture":



Considering the following portion of assembly, describe the corresponding VLIW code:

```
Loop:   LD F0, 0(R1)
        LD F6, 8(R1)
        LD F10, 16(R1)
        LD F14, 24(R1)
        FADD F4, F0, F2
        FADD F8, F6, F0
        FADD F12, F10, F14
        FADD F16, F14, F2
        SD F4, 0(R1)
        SD F8, 8(R1)
        ADD R1, 32
        SD F12, -16(R1)
        BNE R1 R2 Loop
        SD F16, -8(R1)
```

How many FLOPS/cycle?

Answer P2

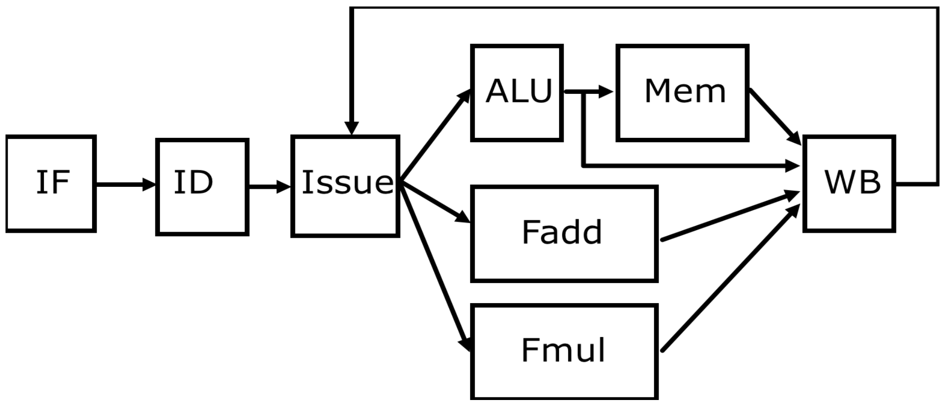
	Int1	Int 2	M1	M2	FP+	FPx

Question 1

Define the concept of static scheduling and describe the main techniques adopted to maximize the Instruction Level Parallelism.

Question 2

Consider the following complex pipeline with a single issue instruction and in which the functional units of the execution stage have different latencies. What type of hazards can occur? What are the possible solutions?



EXTRA

Describe (the answer has to be effectively supported) a 1-BHT and a 2-BHT able to execute the following assembly code (R0 is set to 2000, R1 is set to 0)

```
LOOP:     LD      F1   0     R0
          ADDD    F2   F1    F1
          ADDI    R1   R1    1
LOOP2:    MULTD   F2   F2    F1
          SUBI    R1   R1    1
          BNEZ    R1   LOOP2
          SUBI    R0   R0    4
          BNEZ    R0   LOOP
```

The obtained result, in terms of miss predictions, is inline with theoretical characteristics of the two predictors? Please effectively support your answer.

Answer